

No Dice, Niels: Deterministic Quantum Statistics from Real-Number Quantum Theory and Null-Shell Causality

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Abstract

This paper develops a deterministic account of quantum statistics by combining two established but rarely unified frameworks: time-symmetric null-shell causality and real-number quantum theory (RNQT). Null-shell causality treats all interactions as completed two-boundary events occurring at lightlike separation, in the spirit of direct-action electrodynamics. RNQT demonstrates that complex Hilbert space structure is representational rather than ontological: quantum theory can be formulated entirely over real vector spaces without changing operational predictions [1]. Here these results are fused into a single picture in which the two “components” of a complex amplitude correspond physically to the retarded and advanced contributions of a null-boundary transaction. Once amplitude squaring is assumed, the geometry of these two legs yields the Born probability law without adjustable parameters. A simple symmetry-derived threshold partitions continuous coupling into deterministic single-event outcomes while preserving ensemble statistics. Bell correlations emerge from boundary-conditioned phase selection, where absorber microstates jointly constrain which emission phases complete. The model is retrocausal only in a kinematic sense: future boundary conditions restrict which past events participate in closed null-shell loops, but no signals propagate outside the light cone. Locality and no-signaling are preserved. The resulting framework yields a deterministic account of quantum probabilities grounded in null geometry and real-number Hilbert-space structure.

Keywords

quantum foundations; real-number quantum theory; retrocausality; null-shell causality; Born rule; Bell correlations; direct-action electrodynamics; determinism

1 Introduction

Quantum mechanics is widely regarded as intrinsically probabilistic. Einstein rejected this view, insisting that physical probabilities should reflect incomplete knowledge of underlying variables rather than fundamental indeterminism. Contemporary formulations appear to contradict this hope, yet two independent developments dramatically shift the landscape.

The first is *real-number quantum theory* (RNQT), sharpened by reconstruction results showing that the full operational content of quantum mechanics can be obtained from real Hilbert spaces without assuming complex structure [1]. In RNQT the imaginary unit does not represent an ontological entity but a two-dimensional internal rotation acting on pairs of real components. Complex amplitudes are therefore representational, not physical.

The second is the direct-action, time-symmetric *null-shell causality* framework, inspired by Fokker’s invariant action and Wheeler–Feynman absorber theory [2, 3]. Here physical interactions occur only at lightlike separation, and observable events correspond to completed two-boundary transactions between emission and absorption. Both retarded and advanced contributions participate symmetrically, and the geometry of these null boundaries determines the interaction strength.

These two developments complement each other. The two real components underlying a complex amplitude in RNQT correspond naturally to the retarded and advanced legs of a null-shell transaction. The Born rule emerges from the geometric overlap of these contributions once amplitude squaring is assumed. Deterministic single-event outcomes arise by partitioning continuous coupling with a natural symmetry threshold. Bell correlations follow from boundary-conditioned phase selection without violating locality or no-signaling.

The aim of this paper is to present this unified account. Section 2 reviews RNQT and its implications for complex structure. Section 3 introduces the null-shell framework. Section 4 derives Born statistics. Section 5 presents the deterministic threshold. Section 6 develops Bell correlations. Section 7 compares the framework to other retrocausal interpretations. Section 8 discusses empirical consequences.

2 Real-Number Quantum Theory

RNQT reconstructs quantum mechanics using real Hilbert spaces by embedding complex numbers into 2×2 real matrices that represent a quarter-cycle rotation in an internal plane [1]. The imaginary unit i corresponds to the matrix

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \tag{1}$$

and a complex amplitude $a + ib$ is encoded as a two-component real vector $(a, b)^T$ on which J acts as a $\pi/2$ rotation.

Standard quantum operations—density matrices, observables, tensor products, and measurement statistics—are reproduced faithfully by restricting dynamics to a subspace of “special symmetric” real matrices closed under a modified real tensor product [1]. Crucially, all measurable predictions remain unchanged: RNQT is operationally equivalent to the usual

complex Hilbert-space formulation. The difference lies in representation: complex numbers become convenient notational devices rather than elements of the underlying ontology.

This shows that complex numbers need not appear in the ontology of a physical theory. They encode relations between pairs of real components, not additional physical degrees of freedom. This reinterpretation is ideally matched to time-symmetric null-shell causality, where two real contributions (retarded and advanced) are already fundamental.

3 Null-Shell Causality

Null-shell causality describes interactions as lightlike contacts between worldlines. Each observable event corresponds to a closed retarded/advanced loop between emission and absorption. This is a modernized extension of direct-action electrodynamics [2, 3], in which fields do not propagate as independent entities but arise from boundary-matching conditions between sources and absorbers.

Let λ_e and λ_a denote emission and absorption phases relative to a polarizer axis θ . The retarded and advanced contributions produce real interaction factors

$$R = \cos(\lambda_e - \theta), \quad A = \cos(\theta - \lambda_a). \quad (2)$$

A transaction completes when these two legs mutually reinforce. Their joint amplitude is

$$\mathcal{A} = RA = \cos(\lambda_e - \theta) \cos(\theta - \lambda_a), \quad (3)$$

and the corresponding weight is

$$W = |\mathcal{A}|^2. \quad (4)$$

This weight represents geometric alignment between the two null boundaries. The two real factors R and A play the role of the two components that RNQT otherwise packages into a single complex amplitude.

4 Emergence of Born Statistics

When absorber phases are uniformly distributed over $[0, 2\pi)$, the detection probability becomes

$$P(\theta|\lambda_e) \propto \int_0^{2\pi} \cos^2(\lambda_e - \theta) \cos^2(\theta - \lambda_a) d\lambda_a. \quad (5)$$

The factor $\cos^2(\lambda_e - \theta)$ can be pulled out of the integral, leaving a constant normalization. After normalization one obtains

$$P(\theta|\lambda_e) = \cos^2(\lambda_e - \theta), \quad (6)$$

which is just Malus's law and the Born rule for polarization.

Born's rule thus emerges directly from null geometry once amplitude squaring is assumed. No additional stochastic postulate is required; the result follows from averaging over absorber boundary states and interpreting the two real components of the amplitude as retarded and advanced legs of a transaction. RNQT guarantees that this two-component real structure is sufficient to reproduce the usual complex-amplitude predictions [1].

5 Deterministic Threshold

The continuous weight (4) can be discretized into deterministic outcomes using a natural symmetry threshold. Define a detection criterion

$$\text{Detect if } |\mathcal{A}| > T_{\text{crit}}. \quad (7)$$

A canonical choice is $T_{\text{crit}} = 0.5$, the midpoint of the normalized coupling range. This value has several attractive features:

- It provides a symmetric partition of alignment space into “more aligned” and “less aligned” regions.
- It is consistent with each leg being “open” when $|\cos| > 1/\sqrt{2}$, giving a product threshold of $(1/\sqrt{2})^2 = 0.5$.
- When averaged over absorber phases, it reproduces the Born $\cos^2$ law for detection probabilities.

Microscopic determinism is restored: outcomes depend solely on the triplet $(\lambda_e, \theta, \lambda_a)$. The apparent randomness arises from lack of knowledge of absorber microstates rather than indeterministic dynamics.

6 Bell Correlations

For a polarization singlet, partners have phases $(\lambda, \lambda + \pi/2)$. Local amplitudes at settings a and b are

$$\mathcal{A}_A(\lambda, a) = \cos(\lambda - a), \quad (8)$$

$$\mathcal{A}_B(\lambda, b) = \sin(\lambda - b). \quad (9)$$

Null-shell completion on both wings yields a hidden-variable density

$$\rho(\lambda|a, b) \propto \cos^2(\lambda - a) \sin^2(\lambda - b). \quad (10)$$

Local deterministic outcomes are taken as the signs

$$A(a, \lambda) = \text{sgn}[\cos(\lambda - a)], \quad B(b, \lambda) = -\text{sgn}[\cos(\lambda - b)]. \quad (11)$$

The joint correlation is

$$E(a, b) = \int_0^{2\pi} \rho(\lambda|a, b) A(a, \lambda) B(b, \lambda) d\lambda, \quad (12)$$

which yields

$$E(a, b) = -\cos(2(a - b)), \quad (13)$$

the standard quantum prediction for polarization singlets and the source of the maximal CHSH violation $S = 2\sqrt{2}$.

Bell’s theorem assumes that the hidden-variable distribution $\rho(\lambda)$ is independent of the measurement settings (a, b) (“measurement independence”). In null-shell causality this assumption fails: advanced contributions from both analyzers constrain which emission phases complete the transaction, so $\rho(\lambda|a, b)$ depends on (a, b) . Measurement independence is relaxed, but locality and no-signaling remain intact because all influences propagate along null shells and no controllable signal can be sent to the past.

7 Relation to Other Retrocausal Models

The present framework shares the time symmetry of the transactional interpretation and the two-state-vector formalism, and it is closely related in spirit to boundary-based retrocausal approaches developed by Price and Wharton [4, 5]. It differs by grounding these ideas in a direct-action principle, with interactions constrained to null separation, and by explicitly leveraging RNQT to interpret complex amplitudes as real two-component structures rather than physical waves.

In contrast to purely formal retrocausal models, the null-shell framework emphasizes a concrete geometric picture: interactions are lightlike transactions connecting emission and absorption events, and probabilities measure the alignment of retarded and advanced contributions. RNQT then provides the mathematical language in which these two contributions appear as the real components underlying conventional complex amplitudes [1].

8 Testability

Null-shell causality predicts deviations from standard quantum expectations when absorber phases are anisotropic or when advanced-leg formation is suppressed. For example, if absorber orientations are partially aligned by external fields, the effective distribution of λ_a is no longer uniform, and the thresholded detection statistics may show small, angle-dependent departures from the ideal $\cos^2$ behavior. Similarly, time-asymmetric optical cavities or lossy environments may preferentially suppress advanced contributions, altering absolute detection rates while preserving normalized correlations.

Experiments with biased absorbers, asymmetric cavities, or engineered environments that selectively modify absorber microstates could therefore probe the framework. Any observed deviations must respect no-signaling, providing a nontrivial constraint on viable implementations.

9 Conclusions

RNQT shows that complex amplitudes need not appear in physical ontology [1]. Null-shell causality provides a natural physical interpretation of the resulting real two-component structure: one component for the retarded leg of an interaction, one for the advanced leg. Born statistics emerge geometrically from the overlap of these contributions. Deterministic outcomes follow from a symmetry threshold that partitions continuous coupling into yes/no

events. Bell correlations arise from boundary-condition constraints on which emission phases complete transactions, without violating locality or enabling signaling.

On this view, quantum randomness is epistemic. It reflects incomplete knowledge of absorber microstates and boundary conditions rather than a fundamental indeterministic law. The dice that Einstein rejected are shadows of half-known boundaries on a null shell, not intrinsic features of Nature.

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Data Availability

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Conflict of Interest

The author declares no conflict of interest.

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